Sabin Baral

Hattiesburg, MS | (601) 307-2765 | Sabin.Baral@usm.edu | LinkedIn | GitHub | Portfolio

Undergraduate Researcher in Polymer Science | High-Throughput Material Research and Automation

EDUCATION

The University of Southern Mississippi — Hattiesburg, MS
Bachelor of Science in Polymer Science and Engineering | Minor: Chemistry | Honors Scholar — **May 2028**
Relevant Coursework: Polymer Rheology, Polymer Mechanics, Polymer Processing, Structure Elucidation, Organic Chemistry
GPA: **4.0/4.0** (President's List, Fall 2024 – Spring 2026)

PUBLICATIONS

Upreti, S.; Xu, L.; Moniruzzaman, M.; Wang, Y.; Adhikari, K.; **Baral, S.**; Patton, D.; Ma, B.; Xu, J.; Li, R.; Zhu, C.; Xia, W.; Gu, X. A Robotic High-Throughput Grid-Search Platform for Mapping Phase Behavior in Triblock Copolymer–Homopolymer Blends. *ACS Nano* **2026**, *20* (24), 17303–17315. DOI: 10.1021/acsnano.6c01299

EXPERIENCE

Undergraduate Researcher and Instrumentation Engineer — **Jul 2025 – Present**
Gu Research Group — *Hattiesburg, MS*

- Developed a Python hardware abstraction layer (HAL) integrating robotic arms, motorized stages, and spectrometers into a closed-loop materials lab
- Fabricated and AFM-characterized **1,000+** thin-film block copolymer samples, improving throughput by **40%**
- Assisted **GISAXS/GIWAXS** experiments at **ALS, NSLS-II**, and performed in-house measurements on **100+** samples
- Designed and fabricated custom sample holders and instrument housings, increasing storage capacity by **500%**

US CMS PURSUE Research Intern — **May 2026 – Jul 2026**
Fermi National Accelerator Laboratory (Fermilab) & Kansas State University — *Batavia, IL & Manhattan, KS*

- Cleanroom-tested **50+** TFPX bare modules, calibrating VDD/IREF and validating silicon stability up to 110 V
- Executed automated pixel calibration (PixelAlive, threshold equalization, noise mapping) using **Ph2_ACF**
- Built a Python GUI for CMS Read-Out Chip testing, reducing setup time by **75%** and eliminating terminal-based workflows

VeriScript — AI Transcript Fraud Detection Platform — **Mar 2026 – May 2026**
Mississippi State Board of Nursing — *Jackson, MS | GitHub*

- Designed a serverless pipeline **(S3→Lambda→Step Functions)** for automated transcript processing
- Built a **14-rule** compliance engine and deployed a React/TypeScript dashboard deployed via AWS CDK

Visiting Research Scholar — **Sep 2025**
Lawrence Berkeley National Laboratory — *Berkeley, CA*

- Used automated AFM to characterize phase behavior across **500+** polymer thin-film samples
- Assembled and optimized a robotic spin-coating and thermal annealing system and authored operating protocols

Undergraduate Research Assistant — **Oct 2024 –May 2025**
Center for Optoelectronics and Devices — *Hattiesburg, MS*

- Fabricated organic solar cells via spin coating, thermal evaporation, and encapsulation techniques across **3** device types
- Characterized device performance and stability through **AFM, TGA**, and **DSC** to investigate degradation mechanisms
- Correlated electrical performance with morphological, thermal, and structural changes to evaluate long-term reliability

SKILLS

- **Materials Characterization:** GISAXS/GIWAXS, AFM, DLS, DSC, TGA, FTIR, UV-Vis, Cleanroom Probe Stations
- **Scientific Computing:** Python, MATLAB, Ph2_ACF, CMSITminiDAQ
- **Design, Prototyping & OS:** AutoCAD, OpenSCAD, Cura, Linux (Pop!_OS, AlmaLinux, CentOS)

HONORS AND AWARDS

SPSE Undergraduate Travel Award — Sep 2026
Honors Keystone Award, USM — Jul 2026
Winner, RoboSoccer Tournament – Google Developer Student Clubs, USM — Sep 2025